

**BOISE
GEMS**

**2026
Front Ensemble
Technique Manual
& Audition Packet**

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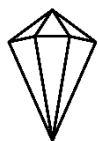
I. INTRODUCTION

Thank you for your interest in the 2026 Boise Gems Front Ensemble!

The exercises in this packet are intended to prepare you for auditions and will be played throughout the entire season, so memorize and perfect them to the best of your ability. If you don't know how to play with four mallets, you can still audition for a keyboard spot! Read the technique manual and do your best. Further instruction will be provided at camps.

The text is just as valuable as the exercises themselves; skipping right to the exercises without reading the rest of the packet **will** cause you to waste your practice time.

Good luck and see you there! If you have any questions, don't hesitate to reach out to me via Slack or email.



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II. AUDITION INFORMATION

Positions available: Marimba (3), Vibraphone (3), Xylophone/Glockenspiel (1-2), Bass Guitar (0-1), Electric Guitar (0-1), Synth (1-2), Rack (1-2)

If you are auditioning for a keyboard instrument, the staff will place you on the keyboard that best suits you. There is no "I am auditioning for marimba," only "I am auditioning for a keyboard spot." You may also be asked if you are willing to play rack or synth.

In-person auditions: You will be playing the packet exercises in an ensemble setting with other Gems auditionees. Bring a printed copy of this packet placed in sheet protectors in a three-ring binder. We are looking for members with:

- A high degree of preparation
- Maturity
- Confidence
- Technical proficiency
- Musical proficiency
- Ensemble skills
- Teachability and humility
- Flexibility, adaptability, willingness to try new things
- A friendly, positive attitude

III. THE GEMS APPROACH

This section is not intended to be the ultimate guide to playing mallet percussion, nor is our technique the only correct way to play mallet percussion. It is simply a brief description of how you will be expected to play as a member of the Boise Gems. Except for a few key differences, much of our approach to mallet playing is derived from Leigh Howard Stevens' *Method of Movement* – consult that book for a significantly more in-depth breakdown.

Posture: The height of your keyboard is essential to playing comfortably and with good technique. The natural keys should be at your beltline, about two inches below your belly button. Stand at your keyboard with your spine straight and arms relaxed by your side, then bend your arms at the elbows. The wrists should be held as low as possible.

Tension: Tension is the #1 issue that leads to injuries in both experienced and inexperienced players alike. It can occur anywhere from the small muscles in your hands and fingers to the large muscles in your back. It can arise from squeezing the mallets too hard (your grip on the mallets should be relaxed yet secure, **never** tight) or trying to play faster than your technique or muscles will allow. If you start to feel tension anywhere in your body as you play, **stop immediately**. Adjust your technique, reduce the tempo, and try again. Remember RBR: **Relax, Breathe, and Rotate**.

Two-mallet technique: The Gems use a back fulcrum, relying on the back two or three fingers to hold the mallet securely in place while the thumb and index finger are simply there for control. This technique allows us to achieve significant power and volume with two mallets. The stroke comes primarily from *wrist flexion* and *extension*, with only slight arm involvement at lower tempos and no arm involvement at higher tempos. As tempos increase (16ths at 200+ bpm), the fulcrum will naturally shift toward the front fingers.



To begin, wrap your pinky and ring finger around the mallet. 1-2 inches of shaft should be sticking out of the back of your fingers. The middle finger rests beside the pinky and ring finger and supports the mallet while the index finger and thumb form a “t” on the mallet shaft with the thumb opposite the outermost joint of the index finger. Ensure that there is space 1) between your thumb and the side of your palm and 2) between your index and middle fingers. Flatten out your wrists so that your palms are facing the floor, then turn them in 10-20°. This is the angle at which you will strike the keys.



Four-mallet technique: We will use Stevens grip for all four-mallet playing. This grip allows us to play intricate passages with many different stroke types, quick interval changes, and a wide range of dynamics.

To hold the mallets, begin with the **outside mallet**. Place it across the first joint of your pinky and ring finger away from your palm, then wrap those two fingers around the mallet with $\frac{1}{4}$ -1 inches of shaft sticking out the back (less experienced players will likely have more due to underdeveloped finger muscles). It should be wedged between your ring finger and just behind the middle knuckle of the middle finger. The mallet should be at a 35-45° angle relative to your forearm.



For the **inside mallet**, lift your index finger and thumb and place the end of the mallet near the center of your palm. Use the tip of your middle finger to anchor it along with the fleshy part of your palm at the base of the thumb. Rest the mallet on the outermost joint of your index finger (commonly referred to as the “perch”), which should be slightly curved, then allow the thumb to rest on the shaft directly opposite that index finger. Avoid pulling the index finger in too far towards your palm or bending your thumb.



Once both mallets are in your hand, turn your wrist so that your thumbnail is facing the ceiling and your palms are facing inward. The mallets should hang effortlessly in the hand and should not be squeezed in any way. The mallet heads should be at the same height – experiment with adjusting your finger position and wrist rotation to achieve this. When you hold your hands in front of you as if ready to play, the mallets are numbered 1-2-3-4 from left to right. **Note** that everyone’s hands are proportioned differently; some may find it more comfortable to *slightly* tuck in the index fingers or to *slightly* wrap the middle finger around the inside mallet rather than using the fingertip to hold the end of the mallet. In addition, hands are slightly asymmetrical and may be most comfortable with slightly different finger positions. Rest assured that minor modifications will not significantly hamper your playing.

Striking the keys: In most cases, you should use as much **velocity** (referring to the speed of the mallet heads into the key) as possible in each stroke without creating tension, even at low dynamics. This allows the core of the mallet to transfer energy into the key, generating a strong sound that projects well. This also allows our dynamic to be controlled solely by height. Ensure that the velocity is still generated by wrist movement, not by pushing with the fingers or arm.

Piston strokes are the fundamental stroke motion used for keyboard instruments. In a piston stroke, your mallets begin at a certain height, move down to strike the keyboard, and move back up to the height of the next stroke. There is no prep or lift – the mallets simply execute the stroke with no wasted motion. Shifts between keys and interval changes between notes should occur during the rebound immediately after striking the key.

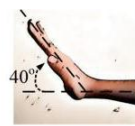
When playing notes significantly isolated in time, a **float stroke** will be used. Strike the key from the initial height and downstroke (stop the rebound just above the key), but do not stop the motion – continue floating the mallets as though you were pulling them back up through molasses. In general, you should lead this motion with the mallet heads and not the wrists. When playing show music, the line between a piston stroke and a float stroke is often blurred. Ultimately, the speed of the rebound depends on what is required by the following notes.

Strive to hit the **center** of the key for each note you play. The only exceptions will be when you are splitting manuals in the same hand (e.g. mallet 1 on the edge of D# and mallet 2 on the center of the next G up) or extremely fast runs.

Four-mallet stroke types: **Double vertical** strokes consist of both mallets in one hand striking the keyboard simultaneously. This is produced by *radial* and *ulnar deviation* with the thumbnail facing the ceiling during the entire stroke. It is very important to ensure that the mallet heads start from the same height to produce an even sound with both mallet heads striking at the exact same time.



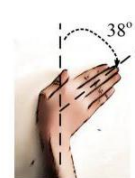
Flexion



Extension



Radial deviation



Ulnar deviation

Single independent strokes, in which a single mallet strikes the keyboard, are produced by rotation of the wrist. Strokes with the inside mallet use *pronation*, while the outside mallet uses *supination*. The mallet that is not playing should remain still and act as the axis of rotation for the other mallet. When playing many inside strokes in quick succession (e.g. scales or a run), the outside mallets should hang loosely at a comfortable middle height to rotate around. This allows us to eliminate slicing while being ready to quickly transition into different stroke types.



Pronation



Supination

Single alternating strokes consist of successive strokes of each mallet in a hand (e.g. 1-2-1-2-1-2). These should be treated as truly individual strokes, mimicking the motions of single independent strokes rather than just rolling the wrist back and forth as in a one-handed roll. Each mallet will rebound to the appropriate middle height to act as the axis of rotation for the mallet that is about to play.

Double lateral strokes occur when two mallets in the same hand strike the keys in quick succession (e.g. outside-inside). At low-to-moderate tempos, these can generally be treated as two successive independent strokes. Similar to single alternating strokes, each mallet should rebound to the height that allows it to act as the axis of rotation for the other mallet. As tempos increase and true independence becomes impossible, double laterals gradually transition to a more relaxed and loose motion.

Advanced stroke types such as one-handed rolls, triple laterals, and accented mallets within double verticals are not covered in this packet and are not expected to be known for auditions. They will be covered in person as needed.

IV. HOW TO PRACTICE

Using a metronome: 99% of your practice should be with a metronome. Technique and note accuracy are meaningless if you can't play rhythms in time. When using a metronome, slow things down until you can play them perfectly, then build the tempo back up in chunks of 4-8 bpm. This creates and preserves good habits, rather than cementing your bad habits by immediately jumping to the highest tempos.

Playing on the floor: It can be very helpful to simplify things by practicing on the floor rather than on a keyboard. This way, you don't have to put any energy into note accuracy or shifting and can place more focus on technique, consistent sound quality, and timing. You can even play along to music you like to make practice more fun.

Being present: Each exercise has a short explainer and a few questions to ask yourself as you play them. Do not ignore these! The point of learning these exercises isn't to learn them, but to **use** them to make your technique, timing, and musicality better. Once you learn the exercises, don't just go on autopilot every time you play them. Be an active participant in your practice. Try to make one thing better with every rep and make every rep count towards improving your playing.

Watching yourself: Especially when working on four-mallet technique, it can be very helpful to place a mirror in front of your keyboard or where you're practicing on the floor so you can see what your mallets are doing. You should also take video recordings of yourself playing the exercises to a metronome. Issues with your technique, balance, and timing may be hard to notice in the moment, but **video does not lie**. Recording yourself and studying these recordings is one of the best ways to get better.

V. EXERCISES

The keyboard exercises in this packet assume a basic knowledge of four-mallet playing. If you are truly just getting started, it may be worthwhile to consult additional resources. These include *Method of Movement* by Leigh Howard Stevens, *Sequential Studies (Book 1)* by Julia Gaines, and [4 Mallet Marimba Technique by Nic Farris](#).

Exercises for rack are intentionally left underwritten to allow for improvisation. Flexibility is an important part of this roles in the front ensemble. Rack auditionees should refer to the drumset parts for a loose guideline. Demonstrate your timing skills and ability to enhance the ensemble's musicality.

Green

Keyboards



This basic scales exercise should be played in all 12 major keys. Natural minor and harmonic minor scales are a bonus. Play with both two mallets and four mallets. Ask yourself the following questions as you practice:

- Are both hands at the same height and velocity?
- Are all notes at the same dynamic? A common tendency is to accent the top and bottom notes of each grouping.
- (when playing with four mallets) Am I using pure wrist rotation rather than pushing with the thumb? Are the inside mallets rotating around the outside mallet head, or are the outside mallets wobbling?

Octaves

Keyboards



Similar to Green, this exercise works on shifting and consistent timing across the keyboard, this time with hands together. Much of the same information from Green applies to Octaves. This exercise should also be practiced as one-handed octaves in Stevens grip.

- Are all shifts started immediately following the previous stroke?
- Does timing remain consistent when moving between naturals and accidentals? i.e., are the wrists moving independently of the elbows, shoulders, and legs?
- (when playing as one-handed octaves) Is the wrist, rather than the arm, the primary driver of the stroke?

Blocks

Keyboards

♩=60-140

The musical score for 'Blocks' for keyboards is presented on a grand staff (treble and bass clefs). It consists of 27 numbered blocks of music. The tempo is marked as ♩=60-140. The score includes various rhythmic patterns, including double vertical strokes and single independent/alternating strokes. The key signature changes throughout the piece, with flats appearing in measures 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, and 27. The score is divided into six systems of five blocks each, with the final system containing only three blocks (25, 26, and 27). The notation includes various accidentals (flats, naturals) and dynamic markings (ff, ff dynamic). The score is written for four mallets, with the first two staves representing the first two mallets and the last two staves representing the last two mallets.

With Blocks, we begin working on four-mallet strokes, namely double vertical and single independent/alternating. Shifting and 8th->16th note rhythm changes are also incorporated.

- Are all four mallets playing at the same height and velocity? Are double verticals, single alternating 8th notes, and single alternating 16th notes the same height and velocity?
- Are double verticals using wrist, not arm (when below ff dynamic)?
- Are single alternating strokes using the non-playing mallet as the axis of rotation, similar to a single independent stroke?
- Does timing remain consistent when moving between naturals and accidentals?
- Is the transition from 8th to 16th notes even and consistent?

Blocks 2

Keyboards

♩=60-140

The musical score for 'Blocks 2' consists of 25 numbered blocks of music, arranged in five rows of five blocks each. The notation is for a single melodic line on a treble clef staff. The tempo is indicated as ♩=60-140. The blocks are numbered 1 through 25. Blocks 1 through 10 are primarily composed of chords and dyads. Blocks 11 through 15 are primarily composed of dyads. Blocks 16 through 19 are primarily composed of dyads. Blocks 20 through 25 are primarily composed of dyads. Blocks 13 and 14 have additional notation below the staff: '13 24 13 24...' and '4 1 3 2 4 1 3 2...' respectively.

Same structure as Blocks but more complex notes, including split manuals and larger shifts. All of the same questions from Blocks also apply here, as well as:

- Is the rhythm affected by interval changes and shifts?
- Are split manual notes the same dynamic as all other notes?

8-16

Keyboards

♩=60-160 for alternating, 60-140 for lateral

1 2 3 4

5 6 7 8

9 10 11 12

13 14 15 16

17 18

19 20 21

This exercise is written as the 1234 permutation, but should be practiced in all lateral (e.g. 4312, 1432) and alternating (e.g. 1423, 3241) permutations. The second half is always the inverse of the first half, so if you're playing 4312 on the way up, you will play 1243 on the way down.

- Is the 8th/16th rhythm transition even and consistent regardless of permutation?
- Do 8th notes use a similar motion to single independents?
- Are 16th notes rebounding to the appropriate height to be used as the axis of rotation for the next mallet to play?
- Do the 16th notes feel just as relaxed as the 8th notes?

Broccoli

Keyboards

$\text{♩} = 60-140$

The musical score consists of 21 measures, each containing a single melodic line on a treble clef staff. The notes are eighth notes, and the measures are grouped into six rows of four measures each, with the final row containing only one measure. Each measure is numbered from 1 to 21. Fingerings are indicated by numbers 1-4 below the notes. The sequence of fingerings across the measures is: 1 (1 2 3 4), 2 (2 3 4 1), 3 (3 4 1 2), 4 (4 1 2 3), 5 (3 2 1 4), 6 (2 1 4 3), 7 (4 3 2 1), 8 (3 2 1 4), 9 (2 1 4 3), 10 (1 4 3 2), 11 (3 2 1 4), 12 (2 1 4 3), 13 (1 2 3 4 2 3 4 1 3 4 1 2 4 1 2 3), 14 (4 3 2 1 3 2 1 4 2 1 4 3 1 4 3 2), 15 (3 2 1 4), 16 (4 3 2 1 3 2 1 4 2 1 4 3 1 4 3 2), 17 (3 2 1 4), 18 (2 1 4 3), 19 (1 2 3 4), 20 (3 2 1 4), 21 (2 1 4 3).

This exercise breaks down all mixed-lateral permutations while demanding focus on even 16th notes over a long period of time.

- Are all four mallets playing at the same height and velocity? A common tendency is to accent the first mallet of each beat or of each hand.
- Is the rhythm even regardless of permutation? Is there a gap, push, or pull before or after the permutation changes?
- Is each hand fully recovering to the top of the stroke after it finishes playing both notes?

Green

Guitars & Drumset

♩=70-180

Guitar

Bass

Bass switch to quarter note around ♩=120

Drumset

The musical score for 'Green' is written for Guitar, Bass, and Drumset. The tempo is marked as ♩=70-180. The Guitar part consists of a series of eighth notes, mostly beamed in pairs, with a final measure containing a whole note and a rest. The Bass part follows a similar pattern of eighth notes, with a note indicating a switch to quarter notes around ♩=120. The Drumset part features a consistent pattern of eighth notes with 'x' marks above them, indicating a specific drum sound, and ends with a star symbol.

Play in all 12 major keys. Natural and harmonic minor scales are a bonus.

Octaves

Guitars & Drumset

♩=60-140

Guitar

Bass

Drumset

The musical score for 'Octaves' is written for Guitar, Bass, and Drumset. The tempo is marked as ♩=60-140. The Guitar part starts in 7/8 time and changes to 4/4 time. The Bass part follows a similar pattern, with a note indicating a switch to quarter notes around ♩=120. The Drumset part features a consistent pattern of eighth notes with 'x' marks above them, indicating a specific drum sound, and ends with a star symbol.

Play in all 12 major keys. Natural and harmonic minor scales are a bonus.

Blocks

Guitar & Drumset

♩=60-140

1 2 3 4 5 6

Guitar

Bass

Drumset

7 8 9 10 11 12 13

G

B

D

14 15 16 17 18 19 20

G

B

D

21 22 23 24 25 26 27

G

B

D

Blocks 2

Guitar & Drumset

♩=60-140

1 2 3 4 5 6

Guitar *Fmaj7* *Am7* *G6* *Bbmaj7* *Amaj7* *Dbm7*

Bass

Drumset 4

7 8 9 10 11 12 13

G *B6* *Emaj7* *Fmaj7* *Am7* *G6* *Bbmaj7* *Amaj7*

B

D 8 4

14 15 16 17 18 19

G *Dbm7* *B6* *Emaj7* *Fmaj7* *Am7* *G6*

B

D 8

20 21 22 23 24 25

G *Bbmaj7* *Amaj7* *Dbm7* *B6* *Emaj7* *Emaj7*

B

D 4 8

8-16

Guitar & Drumset

♩=60-160

Guitar

Bass

Drumset

1 2 3

4 5 6

G B D

7 8 9

G B D

This musical score is for guitar and drumset, spanning measures 1 through 9. The tempo is marked as ♩=60-160. The score is organized into three systems. The first system (measures 1-3) features a guitar part with a treble clef and a key signature of two flats (Bb and Eb), and a bass part with a bass clef and the same key signature. The guitar part includes fingerings 1, 2, and 3. The drumset part is shown on a single staff with a double bar line and a snare drum icon. The second system (measures 4-6) includes guitar (G), bass (B), and drumset (D) parts. The guitar part continues with fingerings 4, 5, and 6. The third system (measures 7-9) also includes guitar (G), bass (B), and drumset (D) parts. The guitar part continues with fingerings 7, 8, and 9. The drumset part uses 'x' marks to indicate specific drum hits on the snare and bass drums.

2

The musical score is divided into four systems, each containing three staves: G (Guitar), B (Bass), and D (Drums/Mallets). The measures are numbered 10 through 21. The notation includes treble and bass clefs, a key signature of one flat (B-flat), and various musical symbols such as eighth notes, sixteenth notes, and rests. The D staff uses 'x' marks to indicate specific mallet strikes on the drum.

Measures 10-12: The first system. Measure 10 starts with a treble clef. Measures 11 and 12 continue the melodic line in the G staff. The B and D staves provide harmonic support.

Measures 13-15: The second system. Measure 13 continues the melodic line. Measure 14 shows a change in the B staff. Measure 15 introduces a key signature change to two flats (B-flat and E-flat) in the G and B staves.

Measures 16-18: The third system. Measure 16 continues the melodic line. Measure 17 shows a change in the B staff. Measure 18 continues the melodic line.

Measures 19-21: The fourth system. Measure 19 continues the melodic line. Measure 20 shows a change in the B staff. Measure 21 ends with a final chord in the G and B staves, followed by a rest in the D staff.

Guitar players should be prepared to play this exercise in any permutation, following the mallet players - the pitches will remain the same, but will be played in a different order.

Broccoli

Guitar & Drumset

♩=60-140

1 2 3 4

Guitar

Bass

Drumset

5 6 7 8

G

B

D

9 10 11 12 13

G

B

D

14 15 16 17

G

B

D

18 19 20 21

G

B

D

Am⁷ B^ø7 C^{maj}7 Dm⁷

Em⁷ F^{maj}7 G⁷ Am⁷ Am⁷ G⁷

F^{maj}7 Em⁷ Dm⁷ C^{maj}7 B^ø7 Am⁷ Am⁷ B^ø7

C^{maj}7 Dm⁷ Em⁷ F^{maj}7 G⁷ Am⁷ Am⁷ G⁷

F^{maj}7 Em⁷ Dm⁷ C^{maj}7 B^ø7 Am⁷ Am⁷